05 · What to control for — and what NOT to (pathmc)

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The business decision behind the statistics. Every effect estimate in every other notebook depends on a choice made *before* any model runs: **which variables to control for** (put in the model / adjust for). Get it wrong and the number is biased no matter how fancy the estimator. The instinct "throw in every column we have" is actively dangerous. This is the notebook that lets you tell a skeptical CMO *exactly why* you controlled for what you did.

1.0.1 The four roles a variable can play — and the rule for each

To "**control for**" (equivalently *adjust for*, *condition on*) a variable W means to hold it fixed when comparing treated vs untreated — e.g. by including it in the regression. Whether that helps or hurts depends entirely on W's causal role between treatment T and outcome Y:

- **Confounder** — a common cause of *both* T and Y (e.g. customer loyalty drives both who gets emailed and how much they spend). It opens a spurious "backdoor" path $T \leftarrow W \rightarrow Y$. -> **Control for it.** Failing to is the classic omitted-variable bias.
- **Mediator** — a variable *on the causal path* $T \rightarrow W \rightarrow Y$ (the effect flows through it). -> **Do NOT control for it** — you'd subtract out part of the very effect you're trying to measure.
- **Collider** — a common *effect* of T and Y ($T \rightarrow W \leftarrow Y$). Conditioning on a collider *opens* a path that was closed, **manufacturing** a correlation that isn't causal. -> **Do NOT.**
- **M-bias collider** — a *pre-treatment* variable that is a collider between two hidden causes. -> **Do NOT** — proof that "just control for everything measured before treatment" is wrong.

The tool that decides this formally is the **backdoor criterion** on the **DAG** (the causal graph): a valid **adjustment set** is any set of variables that blocks every backdoor path from T to Y *without* including a mediator or a descendant of T. `pathmc` reads the graph and enumerates the valid sets for us.

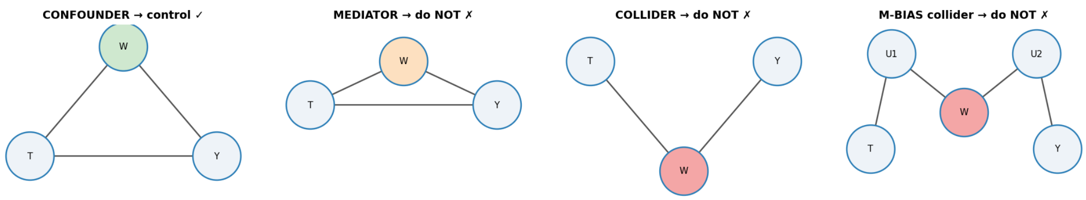
1.0.2 What this notebook does

(a) a **gallery** of the four structures with the correct action; (b) `pathmc` enumerates the admissible adjustment set and flags colliders straight from the DAG; (c) we show the bias of each wrong choice **in euros**; (d) we bound the residual risk from an *unobserved* confounder with a sensitivity contour and an **E-value** (the minimum confounder strength that could explain the effect away).

7-step contract, with **identification** (which variables make the effect recoverable) — not estimation — as the whole point.

1.1 1b · The four structures — a control-or-not gallery

Every "should I control for W?" question reduces to *what role W plays* between treatment T and outcome Y:



- **Confounder** (common cause of T and Y): **control** — it opens a backdoor path.
- **Mediator** (on the causal path T->W->Y): **do not** — you'd remove part of the effect you want.
- **Collider** (common effect of T and Y): **do not** — conditioning *opens* a spurious path.
- **M-bias collider** (pre-treatment, common effect of two latents that separately cause T and Y): **do not** — even a *pre-treatment* variable can be a trap. "Adjust for everything pre-treatment" is wrong.

1.2 2 · Simulate a ground truth

`email -> spend` with a **true effect of €6**. Around it: **`loyalty`** is a **confounder** (drives both email and spend -> must control); **`responded`** is a **collider** (caused by both email and spend -> tempting but poisonous); **`opened_email`** is a post-treatment descendant.

```
TRUE effect of email on spend = €6.0

   email  loyalty  opened_email  responded      spend
0    1.0  1.101262           1.0        0.0  27.705363
1    0.0  0.338431           0.0        0.0  17.425501
2    0.0 -0.539972           0.0        0.0   9.602566
3    0.0 -1.260242           0.0        0.0   5.056443
4    1.0 -1.894621           0.0        0.0   7.175152
```

1.3 3 · Identify — let the DAG choose the adjustment set

The **backdoor criterion**: an admissible set Z blocks every backdoor path T->Y and contains no descendant of T. `pathmc` enumerates admissible sets and flags colliders straight from the graph — **no data needed**, because identification is about structure, not numbers.

```
Admissible adjustment set(s) for email -> spend: [{'loyalty'}]
Identifiable: True

control for {'loyalty'}:  OK - no collider opened
control for {'loyalty', 'responded'}:  'responded' is a collider between 'email'
and 'spend'. Conditioning on it may open a spurious path and introduce bias.
```

```
control for {'opened_email', 'loyalty'}:  OK - no collider opened
control for {}:  OK - no collider opened
```

(Read the printout above as the machine doing the reasoning: it names `{loyalty}` as the one adjustment set that closes the backdoor, confirms the effect is identifiable, and — critically — *warns* when a candidate set includes `responded`, because that variable is a collider. This is the DAG logic from the intro, applied automatically.)

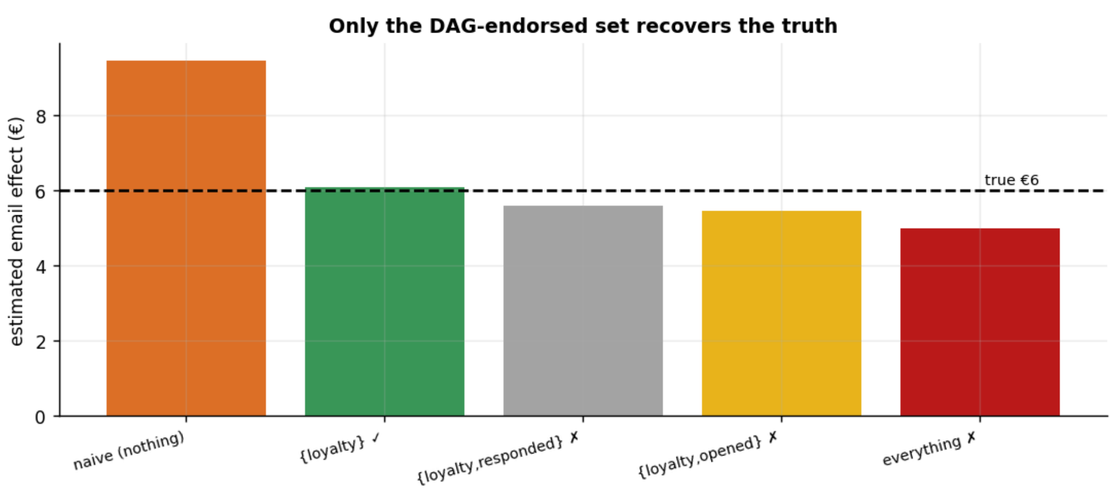
1.4 4–5 · Estimate & Validate — every control choice, in €

Now we make the abstract warnings concrete and expensive. We estimate the same email->spend effect **five ways**, each time controlling for a different set of variables, and compare to the known true €6:

- **nothing** (naive) — leaves the loyalty confounding in, so it's biased *up*;
- **{loyalty}** — the DAG-endorsed set; should land on €6;
- **{loyalty, responded}** — adds a *collider*, which opens a spurious path;
- **{loyalty, opened}** — adds a *post-treatment descendant*, another no-no;
- **everything** — the "throw it all in" instinct, which stacks the errors.

The punchline is that **the sign and rough size of each bias is predictable from the graph *before* you run anything**: omitting a confounder biases up; conditioning on a collider or a post-treatment variable biases in the direction of the association it induces. Controlling for more is *not* safer.

```
naive (nothing)            € 9.45   (error +3.45)
{loyalty} [ok]                € 6.09   (error +0.09)
{loyalty,responded} [x]      € 5.58   (error -0.42)
{loyalty,opened} [x]         € 5.45   (error -0.55)
everything [x]               € 4.97   (error -1.03)
```



How to read this. Only the green bar — `{loyalty}`, the set the DAG endorsed — lands on the true €6. The naive bar overshoots (confounding left in); adding the collider `responded` or the hidden collider `opened_email` (see next) *pulls the estimate away* even though we already had the right confounder; and "everything" is the worst of all. The moral for a practitioner is blunt: **the model can't tell you it's biased — the bars all look like plausible numbers.** The only defence is choosing the control set from the causal graph *before* looking at the estimates, which is precisely what makes the answer defensible to a skeptic.

1.4.1 When the machine is wrong — falsify the DAG against the data

Look back at Step 3: pathmc **green-lit** `{opened_email, loyalty}` ("OK — no collider opened"), yet the bar above shows that exact set is **biased** (−€0.55). That is not a pathmc bug — it is the single most important caveat about DAG-based methods: **the machine reasons only about the graph *we drew.*** Our graph declares `opened_email ~ email`, but the true process has a **hidden engagement trait** driving both `opened_email` and `spend` — making `opened_email` a collider pathmc cannot see, so conditioning on it opens `email -> opened_email <- trait -> spend`.

You can catch this *without* knowing the hidden trait: every DAG makes **testable conditional-independence claims**. Our declared graph implies `opened_email    spend | {email, loyalty}` — so test it against the data:

```
Declared-DAG conditional-independence claims tested: 4  ·  violations: 1
           x     y conditioning_set      p_value
opened_email spend   email, loyalty 1.314202e-19
```

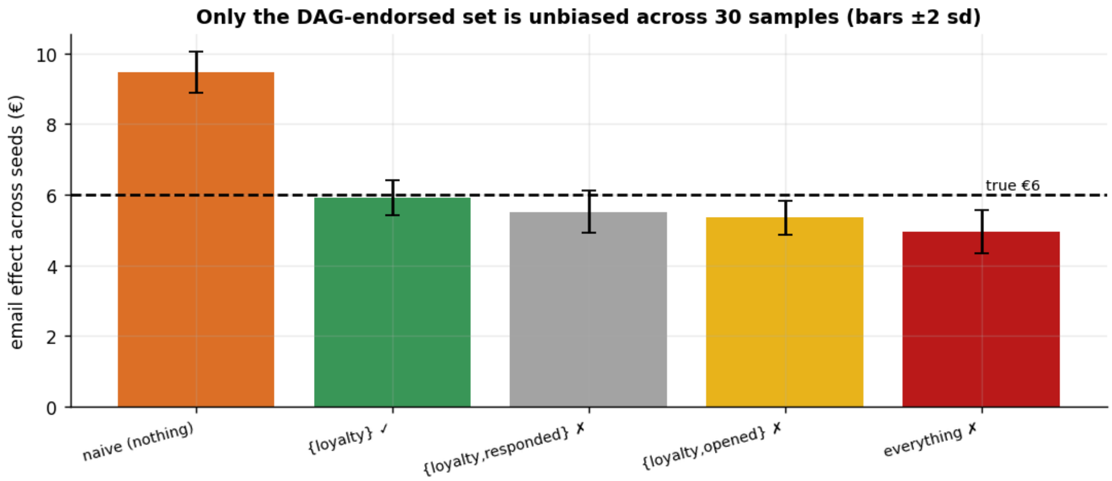
The data **rejects** `opened_email    spend | {email, loyalty}` (p ~ 1e-19): a testable implication of the graph we drew is false, so the graph is misspecified. *That* is why `{opened_email, loyalty}` comes out biased even though pathmc approved it — the tool was faithful to a wrong graph. **The lesson: a causal DAG is an assumption. Let the machine pick the adjustment set from the graph, but *falsify the graph against the data* (here, `test_implications`) before you trust its verdict.**

1.4.2 5b · Recovery across many seeds — unbiased *and* calibrated?

A single sample can land a little off; the real test of an estimator is whether it is **centred on the truth over repeated samples** (unbiased, not just lucky here) and whether its intervals **cover** the truth at their stated rate. We refit on many fresh samples and check both.

```
naive (nothing)            mean € 9.47  bias +3.47  sd 0.30
{loyalty} [ok]                mean € 5.92  bias -0.08  sd 0.25
{loyalty,responded} [x]      mean € 5.51  bias -0.49  sd 0.31
{loyalty,opened} [x]         mean € 5.35  bias -0.65  sd 0.25
everything [x]               mean € 4.96  bias -1.04  sd 0.31

The DAG-endorsed set is centred on the truth every time; the collider / over-
control sets carry a PERSISTENT bias
(a fixed offset that does NOT shrink across seeds, unlike sampling noise) -
exactly what a single sample can hide.
```

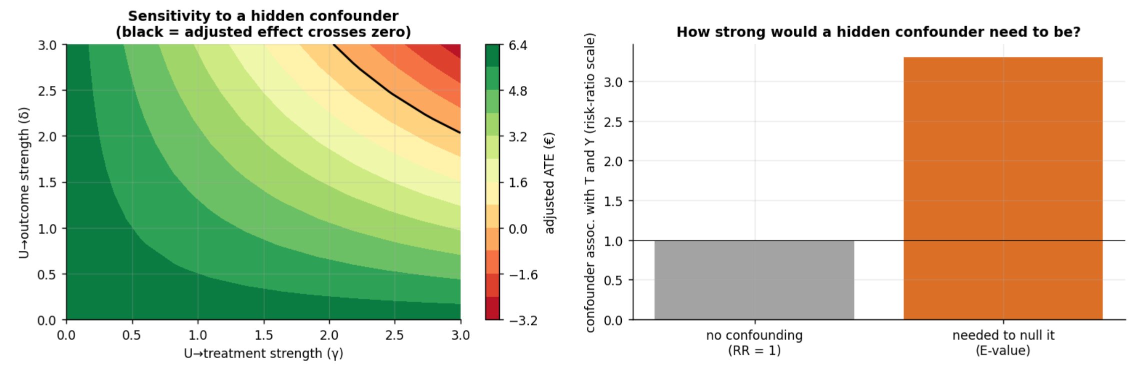


1.5 6 · Decide, in euros — the stakes of the control choice, with a robustness statement

Two parts. **(a) The euro stakes.** Getting the control set wrong isn't academic: budgeting a campaign on the naive (confounded) estimate books *phantom* incremental revenue, which we price out below across a realistic campaign — the concrete cost of skipping the DAG. **(b) The robustness statement.** Identification is untestable, so we bound the residual risk: how strong would an *unobserved* confounder have to be to overturn the conclusion? `pathmc.sensitivity` gives the tipping surface (with the grid widened so the zero-crossing is actually reachable), and the **E-value** summarises it as a single **risk-ratio-scale** number — which is why the euro effect must *not* be plotted on the same axis.

```
Observed adjusted ATE €6.09 (true €6.0)  ·  E-value ~ 3.30
An unmeasured confounder would need ~3.3x association (risk-ratio scale) with
BOTH email and
spend to explain the effect away - ask the domain expert whether anything that
strong is plausible.

Euro stakes: budgeting on the naive €9.45 instead of the DAG-endorsed €6.09
overstates incremental
revenue by €168,031 across a 50,000-customer campaign - phantom lift that would
justify overspending on the email.
control model convergence: max r-hat 1.000 - min ESS 5174 - divergences 0
```



1.6 6b · The stakes on real data (LaLonde) — good vs bad adjustment

This isn't an academic worry. On the famous **LaLonde** job-training data (fetched from a public URL — you provide nothing), the *right* control set recovers the known experimental effect (~$1,800) while a careless one does not. It is the canonical demonstration that *what you adjust for* can flip the sign of your conclusion. Gated on `CMP_REAL=1`.

```
Real-data section skipped. Set CMP_REAL=1 and re-run to fetch the LaLonde
dataset.
```

1.7 7 · Caveats

- **The graph is an assumption.** `pathmc` gives the correct set *for the DAG you drew.* A missing or reversed edge changes the advice — the DAG must encode defensible domain knowledge.
- **Colliders hide in "obvious" controls.** "engaged", "responded", "opened", "clicked" are almost always post-treatment; excluding them feels wrong to practitioners and is exactly right.
- **Front-door as a fallback.** If a T–Y confounder is genuinely unmeasured but a fully-mediating measured variable exists, the front-door adjustment can still identify the effect — a tool to keep in reserve.
- **Sensitivity != proof.** A high E-value means "robust to plausible hidden confounding," not "unconfounded." Pair it with the expert's judgement about what could be missing.